

# Discrete Homework 3

Michael Joseph Ellis

September 28, 2025

## 1. Determine truth values

Throughout,  $x$  denotes an arbitrary object. We interpret  $\subset$  as “proper subset” and  $\subseteq$  as “subset (not necessarily proper).”

- (a)  $x \in \{x\}$     **True.**

By definition, the singleton  $\{x\}$  has exactly one element, namely  $x$ .

- (b)  $\{x\} \subseteq \{x\}$     **True.**

Every set is a subset of itself (reflexivity of  $\subseteq$ ). Since the two sets are equal,  $\{x\} \subseteq \{x\}$  holds.

- (c)  $\{x\} \in \{x\}$     **False.**

The only element of  $\{x\}$  is  $x$  (not the set  $\{x\}$  itself).

- (d)  $\{x\} \in \{\{x\}\}$     **True.**

The set  $\{\{x\}\}$  has exactly one element, namely  $\{x\}$ .

- (e)  $\emptyset \subseteq \{x\}$     **True.**

The empty set is a subset of every set.

- (f)  $\emptyset \in \{x\}$     **False in general (true only if  $x = \emptyset$ ).**

If  $x \neq \emptyset$ , then the sole element of  $\{x\}$  is  $x$ , so  $\emptyset \notin \{x\}$ . (Example: with  $x = 1$  we have  $\emptyset \notin \{1\}$ .)

## 2. Enrollment counts and Venn diagram

Let  $L$  = Linear Algebra,  $C$  = Calculus,  $D$  = Discrete Math. Given:

$$|L| = 60, \quad |C| = 54, \quad |D| = 48, \quad |L \cap C| = 30, \quad |L \cap D| = 24, \quad |C \cap D| = 20, \quad |L \cap C \cap D| = 5.$$

Region counts (pairwise numbers include the triple):

$$\text{only } L : \quad 60 - 30 - 24 + 5 = 11,$$

$$\text{only } C : \quad 54 - 30 - 20 + 5 = 9,$$

$$\text{only } D : \quad 48 - 24 - 20 + 5 = 9,$$

$$L \cap C \text{ only: } 30 - 5 = 25,$$

$$L \cap D \text{ only: } 24 - 5 = 19,$$

$$C \cap D \text{ only: } 20 - 5 = 15,$$

$$L \cap C \cap D : \quad 5.$$

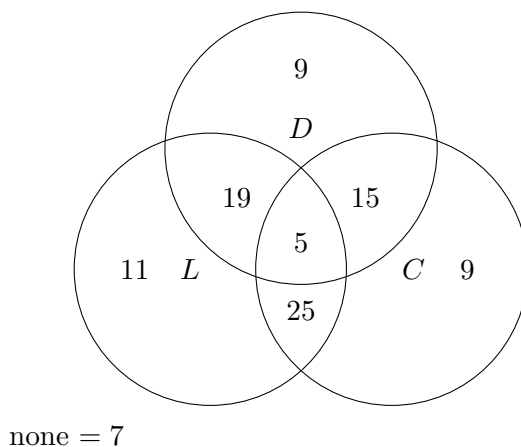
Students in at least one course:

$$11 + 9 + 9 + 25 + 19 + 15 + 5 = 93.$$

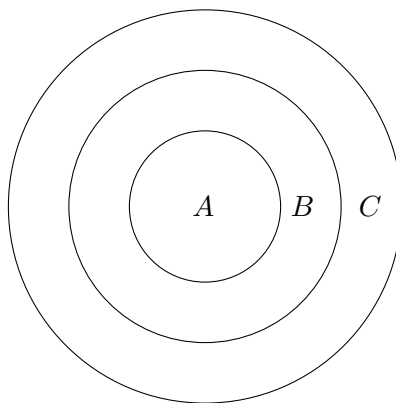
Hence the number who take none is  $100 - 93 = \boxed{7}$ .

**Check (inclusion–exclusion):**

$$|L \cup C \cup D| = 60 + 54 + 48 - 30 - 24 - 20 + 5 = 93 \Rightarrow \text{none} = 100 - 93 = 7.$$



### 3. Diagram for $A \subseteq B$ and $B \subseteq C$



The picture shows  $A$  entirely inside  $B$ , and  $B$  entirely inside  $C$ .

### 4. From $A \subseteq B$ and $B \subseteq C$ to $A \subseteq C$

#### (a) Element-chasing proof

*Proof.* Assume  $A \subseteq B$  and  $B \subseteq C$ . Let  $x \in A$  be arbitrary. Then  $x \in B$ , and therefore  $x \in C$ . Since every element of  $A$  lies in  $C$ , we have  $A \subseteq C$ .  $\square$

#### (b) Truth-table proof

Let  $P$  denote “ $x \in A$ ,”  $Q$  denote “ $x \in B$ ,” and  $R$  denote “ $x \in C$ .” Then  $A \subseteq B$  is  $P \rightarrow Q$  and  $B \subseteq C$  is  $Q \rightarrow R$ . We verify  $(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$  is always true.

| $P$ | $Q$ | $R$ | $P \rightarrow Q$ | $Q \rightarrow R$ | $(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$ |
|-----|-----|-----|-------------------|-------------------|----------------------------------------------------------------------------|
| $T$ | $T$ | $T$ | $T$               | $T$               | $T$                                                                        |
| $T$ | $T$ | $F$ | $T$               | $F$               | $T$                                                                        |
| $T$ | $F$ | $T$ | $F$               | $T$               | $T$                                                                        |
| $T$ | $F$ | $F$ | $F$               | $T$               | $T$                                                                        |
| $F$ | $T$ | $T$ | $T$               | $T$               | $T$                                                                        |
| $F$ | $T$ | $F$ | $T$               | $F$               | $T$                                                                        |
| $F$ | $F$ | $T$ | $T$               | $T$               | $T$                                                                        |
| $F$ | $F$ | $F$ | $T$               | $T$               | $T$                                                                        |

Since the final column is  $T$  in every row, the statement

$$(P \rightarrow Q) \wedge (Q \rightarrow R) \rightarrow (P \rightarrow R)$$

is true for all possible truth values of  $P, Q, R$ . Interpreting  $P, Q, R$  as “ $x \in A$ ,” “ $x \in B$ ,” and “ $x \in C$ ,” this means that whenever  $x \in A$  implies  $x \in B$  and  $x \in B$  implies  $x \in C$ , then  $x \in A$  implies  $x \in C$ . Therefore  $A \subseteq C$ .

## 5. Expressing $A \cap B$ using only $\cup$ and set difference

**Claim.** For all sets  $A, B$ ,

$$A \cap B = A \setminus (A \setminus B).$$

*Proof.* Let  $x$  be arbitrary.

( $\Rightarrow$ ) If  $x \in A \cap B$ , then  $x \in A$  and  $x \in B$ . Hence  $x \notin A \setminus B$ , so  $x \in A \setminus (A \setminus B)$ .

( $\Leftarrow$ ) If  $x \in A \setminus (A \setminus B)$ , then  $x \in A$  and  $x \notin (A \setminus B)$ . The latter means it is not the case that  $x \in A$  and  $x \notin B$ ; since already  $x \in A$ , we must have  $x \in B$ . Therefore  $x \in A \cap B$ .

Both containments hold, so  $A \cap B = A \setminus (A \setminus B)$ .  $\square$

(Optional symmetric/union form.) Also

$$A \cap B = (A \cup B) \setminus ((A \setminus B) \cup (B \setminus A)),$$

but the simpler  $A \setminus (A \setminus B)$  already uses only the allowed operations.

## 6. Functions $\mathbb{Z} \rightarrow \mathbb{Z}_{>0}$ with prescribed properties

Below, the codomain is the positive integers  $\mathbb{Z}_{>0} = \{1, 2, 3, \dots\}$ .

### (a) One-to-one but not onto

Define  $b : \mathbb{Z} \rightarrow \mathbb{Z}_{>0}$  by

$$b(n) = \begin{cases} 2n, & n > 0, \\ 1, & n = 0, \\ -2n + 1, & n < 0, \end{cases}$$

which is a bijection (see part (c)). Then

$$f(n) = 2b(n)$$

maps  $\mathbb{Z}$  injectively into the even positive integers, so  $f$  is one-to-one and not onto (it misses every odd number).

### (b) Onto but not one-to-one

Define

$$g(n) = |n| + 1.$$

For any  $m \in \mathbb{Z}_{>0}$ , choose  $n = m - 1$  to get  $g(n) = m$ ; hence  $g$  is onto. But  $g(m - 1) = g(1 - m) = m$ , so distinct inputs can share the same value;  $g$  is not one-to-one.

### (c) One-to-one and onto

Define

$$h(n) = \begin{cases} 2n, & n \geq 1 \quad (\text{positives map to even numbers}), \\ 1 - 2n, & n \leq 0 \quad (\text{nonpositives map to odd numbers}). \end{cases}$$

*Why this is a bijection.* The two images are disjoint and cover all positives:

$$\{2, 4, 6, \dots\} \cup \{1, 3, 5, \dots\} = \mathbb{Z}_{>0}.$$

On each branch the formula is injective, and the branches don't collide (evens vs odds).

*Explicit inverse (onto + injective in one shot).* Given  $m \in \mathbb{Z}_{>0}$ ,

$$h^{-1}(m) = \begin{cases} m/2, & m \text{ even}, \\ (1 - m)/2, & m \text{ odd}. \end{cases}$$

So every positive integer is hit exactly once, i.e.,  $h$  is one-to-one and onto.

### (d) Neither one-to-one nor onto

Define the constant map

$$k(n) = 1.$$

It is not onto (only 1 is hit) and not one-to-one (all inputs share the same value).

## 7. Bijections $\mathbb{R} \rightarrow \mathbb{R}$ ? Prove or disprove

(a)  $f(x) = 2x + 1$ . **Bijection.**

*Injective:* If  $2x_1 + 1 = 2x_2 + 1$  then  $x_1 = x_2$ . *Surjective:* Given  $y \in \mathbb{R}$ , take  $x = (y - 1)/2$  and  $f(x) = y$ . Inverse:  $f^{-1}(y) = \frac{y - 1}{2}$ .

(b)  $f(x) = x^2 + 1$ . **Not a bijection.**

*Not injective:*  $f(1) = f(-1) = 2$ . *Not surjective:*  $f(\mathbb{R}) = [1, \infty)$ , so no  $x$  with  $f(x) = 0$ .

(c)  $f(x) = x^3$ . **Bijection.**

*Injective:* If  $x_1^3 = x_2^3$  then  $x_1 = x_2$ . *Surjective:* Given  $y \in \mathbb{R}$ ,  $x = \sqrt[3]{y}$  satisfies  $f(x) = y$ . Inverse:  $f^{-1}(y) = \sqrt[3]{y}$ .

(d)  $f(x) = \frac{x^2 + 1}{x^2 + 2}$ . **Not a bijection.**

Note  $f(x) = 1 - \frac{1}{x^2 + 2}$ . As  $x$  ranges over  $\mathbb{R}$ ,  $x^2 + 2 \in [2, \infty)$ , so

$$f(\mathbb{R}) = \left[ \frac{1}{2}, 1 \right).$$

Hence  $f$  is *not* surjective onto  $\mathbb{R}$ . Also  $f(1) = f(-1)$ , so it is not injective.

## 8. Example with $f \circ g$ bijective but $g$ not onto and $f$ not one-to-one

Let  $A = \{1, 2\}$ ,  $B = \{1, 2, 3\}$ ,  $C = \{1, 2\}$ .

Define  $g : A \rightarrow B$  by  $g(1) = 1$ ,  $g(2) = 2$ . Then  $g$  is not onto (it misses 3).

Define  $f : B \rightarrow C$  by  $f(1) = 1$ ,  $f(2) = 2$ ,  $f(3) = 2$ . Then  $f$  is not one-to-one (since  $f(2) = f(3)$ ).

Composition:  $(f \circ g)(1) = f(1) = 1$  and  $(f \circ g)(2) = f(2) = 2$ , so  $f \circ g : A \rightarrow C$  is the identity on  $\{1, 2\}$ , hence a bijection.

## 9. If $g : A \rightarrow B$ and $f : B \rightarrow C$ are injective, then $f \circ g$ is injective

*Proof.* Assume  $g : A \rightarrow B$  and  $f : B \rightarrow C$  are injective. Let  $x, y \in A$  and suppose  $(f \circ g)(x) = (f \circ g)(y)$ . Then  $f(g(x)) = f(g(y))$  with  $g(x), g(y) \in B$ . Since  $f$  is injective,  $g(x) = g(y)$ . Since  $g$  is injective,  $x = y$ . Thus for all  $x, y \in A$ ,  $(f \circ g)(x) = (f \circ g)(y) \Rightarrow x = y$ , so  $f \circ g$  is injective.  $\square$